

MINNIPA, Australia

WMO: 956620

Lat: 32.8428S

Lon: 135.1514E

Elev: 166

StdP: 99.35

Time Zone: 9.50 (AUA)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 4.4	(c) 5.3	(d) -8.4	(e) 1.9	(f) 25.6	(g) -6.2	(h) 2.3	(i) 26.5	(j) 12.2	(k) 13.9	(l) 11.1	(m) 14.5	(n) 3.4	(o) 130	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 15.5	(c) 39.8	(d) 18.4	(e) 37.6	(f) 17.7	(g) 35.4	(h) 17.2	(i) 21.1	(j) 30.0	(k) 20.1	(l) 30.4	(m) 19.1	(n) 29.6	(o) 6.2	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.9	(b) 14.0	(c) 22.9	(d) 17.4	(e) 12.7	(f) 22.0	(g) 16.0	(h) 11.6	(i) 21.1	(j) 61.9	(k) 29.0	(l) 58.1	(m) 30.2	(n) 54.9	(o) 29.1	(p) 24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.2	10.0	9.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			2.3	43.8	0.9	1.6	1.6	44.9	1.1	45.9	0.6	46.8	-0.1	48.0
			1.6	22.7	0.9	1.5	1.0	23.7	0.5	24.6	0.0	25.4	-0.6	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	24.0	23.6	21.2	18.4	14.7	11.9	11.3	12.1	14.5	17.4	20.0	21.8
	DBStd	5.81	4.69	4.53	4.26	3.65	2.77	1.97	2.02	2.66	3.70	4.26	4.68	4.76
	HDD10.0	19	0	0	0	0	0	4	8	6	2	0	0	0
	HDD18.3	1034	3	4	16	42	118	194	218	194	127	71	33	15
	CDD10.0	2774	434	380	347	253	147	61	48	72	137	231	299	366
	CDD18.3	745	178	151	104	46	7	1	0	1	12	43	83	122
	CDH23.3	9602	2401	1866	1187	439	33	1	1	8	166	627	1162	1712
	CDH26.7	5043	1407	1041	580	153	3	0	0	1	45	268	596	950
(14) Wind	WSAvg	5.1	5.9	5.6	5.3	4.8	4.4	4.4	4.7	4.8	5.0	5.3	5.4	5.6
(15) Precipitation	PrecAvg	313	12	16	17	18	28	44	42	41	32	23	18	22
	PrecMax	530	78	60	64	47	74	92	88	81	74	83	48	100
	PrecMin	142	0	0	0	1	2	13	13	4	2	0	1	0
	PrecStd	73	16	15	19	13	16	21	18	22	20	22	12	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	41.4	38.6	34.2	26.6	21.4	21.2	24.5	31.1	35.5	39.2	41.5
		MCWB	19.4	19.8	17.8	16.6	14.1	12.5	11.7	12.4	14.5	15.8	17.0	18.0
	2%	DB	40.2	38.6	35.3	30.6	23.8	18.7	18.5	21.8	27.6	32.5	35.9	38.2
		MCWB	18.7	18.7	17.1	15.1	13.1	11.3	10.9	11.8	13.4	14.8	16.6	17.4
	5%	DB	37.3	36.1	32.6	28.0	21.6	17.0	16.7	19.6	25.0	29.6	32.9	35.4
		MCWB	18.0	18.2	16.7	14.5	12.6	11.2	10.9	11.2	12.8	14.0	15.7	16.6
	10%	DB	34.3	33.0	29.9	25.6	19.6	15.6	15.2	17.3	22.2	26.8	30.0	32.3
		MCWB	17.6	17.6	16.2	13.9	12.2	11.0	10.4	10.5	12.1	13.2	15.1	16.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.7	21.3	18.7	16.2	15.0	14.0	14.4	16.2	17.6	20.1	21.3
		MCDB	30.9	32.0	28.5	23.6	21.1	18.4	16.0	20.3	25.1	27.5	27.7	28.4
	2%	WB	20.7	21.3	19.7	17.1	15.1	13.5	12.6	13.3	14.9	16.2	18.2	19.6
		MCDB	32.0	30.2	27.6	24.5	19.3	15.7	15.8	17.6	23.0	26.4	29.8	29.2
	5%	WB	19.7	20.1	18.5	16.1	14.4	12.6	11.9	12.4	13.8	15.2	17.2	18.4
		MCDB	32.3	29.9	26.8	23.4	18.3	14.8	14.9	16.7	21.8	25.3	29.1	29.6
	10%	WB	18.7	18.9	17.5	15.3	13.5	11.9	11.1	11.6	12.9	14.3	16.2	17.3
		MCDB	30.8	29.2	26.2	22.0	17.6	14.3	14.1	15.9	20.0	23.7	27.1	29.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	14.6	13.2	11.7	9.2	7.9	8.2	10.0	12.4	14.4	15.0	15.1
		MCDBR	19.2	17.8	16.4	14.7	11.5	9.5	10.0	12.8	16.5	18.5	18.5	19.1
		MCWBR	5.2	4.9	4.7	4.7	4.7	4.9	5.2	5.8	6.0	6.1	5.7	5.6
	5% WB	MCDBR	15.3	13.9	13.1	11.5	8.9	7.3	8.5	10.4	14.3	15.5	15.9	15.0
		MCWBR	5.2	5.0	4.7	4.8	4.8	4.9	5.3	5.8	6.1	6.0	5.7	5.7
(40) Clear-Sky Solar Irradiance	taub		0.339	0.339	0.320	0.312	0.300	0.292	0.293	0.300	0.327	0.337	0.341	0.330
	taud		2.518	2.536	2.571	2.579	2.595	2.603	2.593	2.560	2.473	2.447	2.468	2.525
	Ebn at Noon		999	979	963	918	879	863	879	918	941	969	990	1011
	Edh at Noon		112	106	96	85	74	69	73	85	104	115	117	112
(44) All-Sky Solar Radiation	RadAvg		7.88	7.08	5.82	4.31	3.12	2.65	2.85	3.77	5.11	6.40	7.29	7.82
	RadStd		0.39	0.38	0.33	0.27	0.14	0.12	0.12	0.22	0.34	0.42	0.34	0.38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+60

Nomenclature: See separate page